

Security Audit Report

Week 1 to Week 4 Practical Task

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Project Area: n8n Automation, Webhooks, AI Workflows, APIs and Database Integration

Audit Period: Week 1 to Week 4

Purpose: Security review of workflows, credentials, webhooks and project files

Executive Summary

This security audit was conducted on the workflows and projects developed during Week 1 to Week 4. The purpose of the audit was to identify exposed credentials, hard coded secrets, insecure webhook configurations and other common security issues.

The audit focused on three main objectives:

1. Verify that API keys, passwords, tokens and other secrets were not hard coded or committed to the project repository.
2. Protect one webhook using authentication through a secret HTTP header.
3. Document the security issues identified and the corrective actions taken.

The audit resulted in the removal or prevention of hard coded credentials, improved credential management and the addition of authentication to a webhook.

Secret and Credential Audit

The workflows were inspected for commonly exposed secrets including:

Gemini API keys
Groq API keys
OpenAI API keys
Supabase keys
Database passwords
Access tokens
Bearer tokens
Webhook secrets
SMTP credentials
Authorization headers

The following areas were specifically reviewed:

HTTP Request nodes
Code nodes
Set nodes
AI Agent prompts
Webhook nodes

Environment variables
n8n Credentials
Project files
Execution logs

Finding

Hard coded credentials present a significant security risk because anyone with access to the workflow file, repository, export or execution data may potentially obtain the credential.

Action Taken

Sensitive credentials were moved to secure credential storage rather than being placed directly inside workflow logic.

n8n Credential Management

n8n Credentials were used for services requiring authentication.

Examples include:

Google Gemini
Groq
Supabase
Gmail
Database connections
Other authenticated APIs

The credentials are referenced by the workflow rather than manually placing secret values inside individual nodes.

Security Improvement

This reduces the likelihood of accidentally exposing credentials when exporting workflows, sharing screenshots or committing workflow files to Git.

Webhook Security Testing

Test : Missing Header

A request was sent without the required authentication header.

Expected: Unauthorized request is rejected.

Result: Request rejected successfully.

Security Improvements

The following improvements were completed during the audit:

Credential protection

Sensitive API credentials are stored securely instead of being directly embedded in workflow logic.

Webhook authentication

A secret authentication header was added to one webhook.

Conclusion

The Week 1 to Week 4 projects were reviewed from a basic application security perspective, with particular attention to credential management, webhook authentication, source code exposure and repository security.

The audit identified areas where sensitive information could potentially be exposed and applied appropriate corrective measures. The most important improvement was the introduction of authentication on a webhook, ensuring that unauthorized requests cannot proceed to sensitive workflow operations.

The final result is a more secure workflow environment with improved credential management and reduced risk of accidental exposure of secrets.